Tommaso Grigoletto

– Postdoctoral fellow –

Birth date:	02/01/1996
Nationality:	Italian
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Research statement

My name is Tommaso Grigoletto and I am a young Italian researcher. I obtained my Ph.D. and Master degree from the Department of Information Engineering, at the University of Padova where I am currently a postdoctoral fellow. My research interest is situated at the intersection of control theory and quantum science and technology and focuses on theoretical and numerical techniques for the analysis and control of complex dynamical systems, incorporating both classical and quantum probability theory. I developed a foundational interest in automation engineering and dynamical systems early on and it was augmented during my Master's by an introduction to quantum physics. My Ph.D. expanded my interest in the fundamental mathematical challenges associated with quantum technologies and made me discover how methodologies from control systems theory can be harnessed to advance this fundamental frontier in physics.

Professional Experience

Research experience

01/07/2025 – 30/06/2026 **Postdoctoral Fellow** University of Padua, Padua, Italy
Project: Developing aggregation methods for network model reduction
Advisors: Prof. Mattia Zorzi and Prof. Francesco Ticozzi

01/11/2023 – 30/06/2025 **Postdoctoral Fellow** University of Padua, Padua, Italy
Project: Approximate reduction methods for classical and quantum networks
Advisor: Prof. Francesco Ticozzi

06/08/2022 – 10/02/2023 **Visiting Scholar** Dartmouth College, Hanover, NH (USA)
- Project: Exact model reduction for continuous-time quantum dynamics
- Advisor: Prof. Lorenza Viola

Teaching experience

Main instructor	Fundamentals of Automation Engineering,	Fall 2025 (48 hrs)
	Bachelor's degree in Engineering Physics,	Fall 2024 (48 hrs)
	Ca'Foscari University of Venice	Fall 2023 (48 hrs)
Teaching assistant	Control Engineering Laboratory,	Spring 2023 (26 hrs)
	Master's degree in Control Systems Engineering	Spring 2022 (26 hrs)
	University of Padua	

Education

01/10/2020 – 30/09/2023 **Ph.D. in Information Engineering**, University of Padua Padua, Italy
Thesis: Exact model reduction for quantum systems
Advisor: Prof. Francesco Ticozzi
Defended March 21st, 2024, **with honors**.

29/09/2018 – 21/07/2020	**M.Sc. in Automation Engineering**, University of Padua, Padua, Italy Thesis: Measurement-based switching control for quantum systems Advisor: Prof. Francesco Ticozzi Final grade: 110/110
28/08/2015 – 16/07/2018	**B.Sc. in Information Engineering**, University of Padua, Padua, Italy Thesis: Control system for a rocket soft landing Advisor: Prof. Mauro Bisiacco Final grade: 106/110
Sep. 2010 – Jul. 2015	**High school**, I.T.I.S. Rossi, Vicenza, Italy Thesis: Control system for a quadcopter Advisor: Prof. Paolo Fumene Ferruglio Final grade: 94/100

Awards, Fellowships and Grants

2026 "Seal of Excellence" (score 92.4%) for the project "IMPAQT: Improved Magic state Purity viA Quantum conTrol" submitted to the **Marie Sklodowska-Curie Action Postdoctoral Fellowship** in conjunction with Prof. Alain Sarlette from Inria Paris;
2022 Gini scholarship, Fondazione Aldo Gini, University of Padua;
2020 Ph.D. scholarship, University of Padua;
2015 Tekne scholarship, Regione Veneto;
2014 Leonardo scholarship, Fondazione Centro Produttività Veneto;
2013 Participated to the Robocup World Cup in Eindhoven.

Seminars and Lectures

Talks at national and international conferences

1. January 26th 2026 – **"Krylov spaces and algebras for efficient simulation of quantum dynamics"**, QCSC Workshop 2025, Trento Italy;
2. September 9th 2025 – **"Exact Model Reduction for Continuous-Time Open Quantum Dynamics"**, Poster presentation at IQIS 2025, Bologna Italy;
3. December 2024 – **"Exact Model Reduction for Discrete-Time Conditional Quantum Dynamics"**, Invited session at IEEE Conference on Decision and Control, Milan;
4. December 2022 – **"Minimal resources for exact simulation of quantum walks"**, IEEE Conference on Decision and Control, Cancun ;

Other talks

1. December 11th 2026 – **"Krylov spaces and algebras for efficient simulation of quantum dynamics"**, QS-NOW Workshop 2026, Asiago Italy;
2. March 13th 2025 – **"Exact Model Reduction for Quantum Dynamics"**, Invited talk at Quantum Nano Seminar - Dartmouth College;
3. January 20th 2025 – **"Applications of quantum conditional expectations"**, Two-hour lecture included in the Ph.D. course "Quantum Probability Information and Dynamics" held by Francesco Ticozzi, University of Padua;
4. January 10th 2025 – **"Quantum conditional expectation and the Takesaki Theorem"**, Two-hour lecture included in the Ph.D. course "Quantum Probability Information and Dynamics" held by Francesco Ticozzi, University of Padua;
5. March 21st 2024 – **"Exact model reduction for quantum systems"**, Ph.D. Thesis defense, University of Padua;
6. September 22nd 2023 –**"Exact model reduction for Quantum Systems: an algebraic approach"**, Invited talk at Inria Paris, Quantic group;
7. June 27th 2023 – **"Exact model reduction for Quantum Systems: an algebraic approach"**, Seminar talk for the quantum lunch at the department of information engineering, University of Padua;
8. April 2022 – **"Algebraic reduction of Hidden Markov Models"**, Seminar talk at the department of information engineering, University of Padua ;

Mentoring

Bachelor Students

2025	Giacomo Girotto Ca'Foscari	"Experiments on a single qubit quantum computer", Supervisor: **T.G.**
2025	Rosanna Mantese Ca'Foscari	"Control of a Boeing v-22 Osprey", Supervisor: **T.G.**, Co-Supervisor: Matthias Pezzutto
2025	Marco Grolla Ca'Foscari	"Control of a rocket soft-landing", Supervisor: **T.G.**, Co-Supervisor: Matthias Pezzutto
2025	Stefano Santello Ca'Foscari	"Analysis and control of a pendulum actuated with reaction wheels", Supervisor: **T.G.** , Co-Supervisor: Matthias Pezzutto
2025	Marco Tesio Ca'Foscari	"Simulation of open quantum systems using quantum computers", Supervisor: **T.G.**
2025	Marco Vanzetto Ca'Foscari	"Parameter estimation for closed quantum systems", Supervisor: **T.G.**
2024	Giuseppe D'Auria UniPD	"Computer based state observers for magnet systems", Supervisor: Damiano Varagnolo, Co-Supervisor: **T.G.**
2024	Jacopo Tomasetig Ca'Foscari	"Modelling and Control of tire-road interaction dynamics", Supervisor: **T.G.**
2024	Alessandra Dal Bello Ca'Foscari	"Tracking visual markers for maglev platforms", Supervisor: **T.G.**, Co-Supervisor: Damiano Varagnolo

Professional skills

Reviewing, Service and Outreach

- Reviewed for various journal and conferences including:
  IEEE Control System Letters (12 papers),
  IEEE Conference on Decision and Control 2025 (2 papers),
  IEEE Conference on Decision and Control 2024 (4 papers),
  IEEE Conference on Decision and Control 2022 (1 paper),
  IEEE Conference on Decision and Control 2021 (1 paper),
  International Journal of control (1 paper),
  Journal of Physics A (1 paper, received the IOP Trusted Reviewer status for an 'Outstanding' 5/5 review report),
  Automatica (3 papers),
  IEEE American control conference 2025 (1 paper).
- Associate editor for ECC 2026.
- Associate editor for IEEE qCCL 2026.
- Founder and main organizer of the Junior Qtech seminar cycle: a postdoc led cycle of seminars aimed at merging the skills of physicists, mathematicians, engineers and chemists of the Padua QTech center.
- Co-organizer with Prof. Ticozzi of the Invited session "New tools for estimation, modeling and control of quantum systems" at the Conference on Decision and Control 2024.

Languages

Mother tongue:	Italian
Other languages:	English ($\simeq$ C1 level)

Programming skills

Basic:	ECDL, CAD design, HTML, JavaScript, PLC programming languages
Intermediate:	Java, C++, Excel
Advanced:	Python, Matlab, LaTeX

Pubblications

*Legend: *=Shared first-authorship*

Journal

[16] M. Cortese, **T. Grigoletto**, F. Ticozzi, and A. Ferrante. "Robust positive model reduction via monotone matrices". In: *IEEE Transaction on Automatic Control* (2025).

[14] **T. Grigoletto**, C. Pellegrini, and F. Ticozzi. "Quantum model reduction for continuous-time quantum filters". In: *Annales Henri Poincaré* (2025).

[13] **T. Grigoletto** and F. Ticozzi. "Model Reduction for Quantum Systems: Discrete-time Quantum Walks and Open Markov Dynamics". In: *IEEE Transactions on Information Theory* (2025).

[11] **T. Grigoletto**, Y. Tao, F. Ticozzi, and L. Viola. "Exact Model Reduction for Continuous-Time Open Quantum Dynamics". In: *Quantum* 9 (2025), p. 1814.

[7] **T. Grigoletto** and F. Ticozzi. "Exact Model Reduction for Discrete-Time Conditional Quantum Dynamics". In: *IEEE Control Systems Letters* (2024).

[6] W. Liang, **T. Grigoletto**, and F. Ticozzi. "Dissipative feedback switching for quantum stabilization". In: *Automatica* 165 (2024), p. 111659.

[4] **T. Grigoletto** and F. Ticozzi. "Algebraic Reduction of Hidden Markov Models". In: *IEEE Transactions on Automatic Control* 68.12 (2023).

[2] **T. Grigoletto** and F. Ticozzi. "Stabilization via feedback switching for quantum stochastic dynamics". In: *IEEE Control Systems Letters* 6 (2021), pp. 235–240.

Conference

[5] M. Peruzzo, **T. Grigoletto**, and F. Ticozzi. "Reconstructing Quantum States from Local Observation: A Dynamical Viewpoint". In: *2024 IEEE 63st Conference on Decision and Control (CDC)*. 2024.

[3] **T. Grigoletto** and F. Ticozzi. "Minimal resources for exact simulation of quantum walks". In: *2022 IEEE 61st Conference on Decision and Control (CDC)*. 2022, pp. 5155–5160.

[1] T. Benciolini*, T. Grigoletto*, and M. Zorzi. "Image compression by means of the multidimensional circulant covariance extension problem–Revisited". In: *2020 59th IEEE Conference on Decision and Control (CDC)*. IEEE. 2020, pp. 280–285.

Under review

[15] **T. Grigoletto**. *Computing time-dependent reduced models for classical and quantum dynamics*. 2025.

[12] **T. Grigoletto**, L. Viola, and F. Ticozzi. *Model reduction for controlled quantum Markov dynamics*. 2025.

[10] M. Peruzzo, **T. Grigoletto**, and F. Ticozzi. *Quantum dynamical tomography*. 2025.

Other references

[9] **T. Grigoletto**. "Exact model reduction for quantum systems". Ph.D. dissertation. University of Padua, 2024.

[8] **T. Grigoletto**, A. Gallina, and G. Pillonetto. *Applied Functional Analysis and Machine Learning*. Lecture notes. 2024.

In preparation

[18] **T. Grigoletto**, A. Sarlette, F. Ticozzi, and L. Viola. *Adiabatic and Algebraic Methods for Constructing Reduced Lindblad Dynamics*. In preparation.

[17] **T. Grigoletto**, M. Casanova, L. Viola, and F. Ticozzi. *Finding weak symmetries of open quantum systems through Riemannian gradient descent*. In preparation.